Day 135 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Unit 9 Test

Standards: 6.G.A.1, 6.G.A.3, 6.SP.A.1, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.4, 6.SP.A.3, 6.SP.B.5d, 6.SP.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 10, 13, 23

Day 135 | Unit 09 | Unit Test

Display, Describe, and Summarize Data

Unit 9 Test

Standards: 6.G.A.1, 6.G.A.3, 6.SP.A.1, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.4, 6.SP.A.3, 6.SP.B.5d, 6.SP.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

Directions

The first 4 questions (20%) review Unit 8. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Sketch a triangle with base 8 cm and perpendicular height 4 cm. Find its area.

02 / Previous unit review | Constructed response

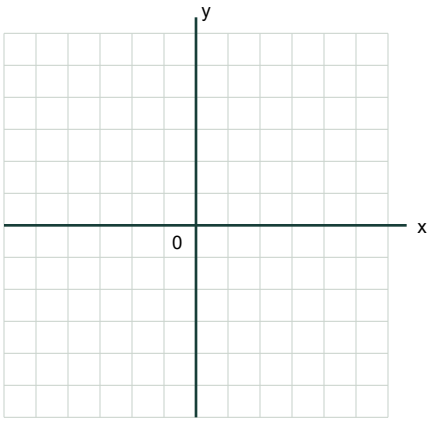
A parallelogram has base 2 m and perpendicular height 3 m. Sketch how it can be rearranged into a rectangle and find its area.

03 / Previous unit review | Constructed response

An L-shaped garden is a 6 m by 6 m rectangle with an 2 m by 2 m corner rectangle removed. Sketch and find the garden area.

04 / Previous unit review | Constructed response

Plot and connect (0,0), (9,0), (9,9), (0,9). Find the side lengths and area.



Label the scale and units.

05 / Constructed response

Is “What are the heights of the plants in our garden?” statistical? Explain.

06 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Report n, units, and mean.

07 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Draw a labeled dot plot. Check the number of dots.

08 / Constructed response

Reading times in minutes are 7, 9, 9, 11, 11, 13. Find the five-number summary and draw a labeled box plot.

09 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Draw a labeled dot plot. Check the number of dots.

10 / Constructed response

Daily reading times in minutes: 12, 15, 14, 19, 12, 12, 6, 15, 8, 5. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

11 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Find Q1, Q3, IQR, and range.

12 / Numeric entry and interpretation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

13 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 7, 7, 9, 9, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

14 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 7, 13, 13, 19, 19, 25. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

15 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 23. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

16 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 6, 11, 11, 16, 16, 21. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

17 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. A second group has every corresponding time 4 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

18 / Data display and numeric entry

A frequency table records books read: 0 books: 1 students; 1: 3; 2: 4; 3: 5. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

19 / Multiple select

A class studies a collection of students. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 2 meters?
B. What is the reading time in minutes of each of the students?
C. What is 2 + 5?
D. How does reading time vary among the students?

20 / Multiple select

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. Select all true statements.

A. The median is the average of the 3rd and 4th ordered values.

B. The median is 10 minutes.

C. The median is always the largest observation.

D. The median is 9 minutes.

Day 135 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Unit 9 Test

Standards: 6.G.A.1, 6.G.A.3, 6.SP.A.1, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.4, 6.SP.A.3, 6.SP.B.5d, 6.SP.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 16 cm²

A = bh/2 = 8 × 4 ÷ 2 = 16. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

2. 6 m²

Cut and shift a corner; A = bh = 2 × 3 = 6.

6.G.A.1 | Constructed response

3. 32 m²

Outer area 36 minus cutout 4 = 32.

6.G.A.1 | Constructed response

4. Sides 9 and 9; area 81 square units

Horizontal length 9 - (0) = 9; vertical length 9 - (0) = 9; area = 9 × 9.

6.G.A.3, 6.G.A.1 | Constructed response

5. Yes; plant heights vary.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. n = 6; minutes; mean 9/2 min

Sum = 27; mean = 27 ÷ 6 = 9/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

7. 8: 1 dot; 9: 2 dots; 10: 2 dots; 11: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

8. (7, 9, 10, 11, 13)

Box runs from 9 to 11, median line at 10, whiskers to 7 and 13.

6.SP.B.4, 6.SP.B.5c | Constructed response

9. 6: 1 dot; 10: 2 dots; 14: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

10. Frequencies 0, 3, 4, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

11. Q1 11, Q3 17; IQR 6 min; range 18 min

Lower half [5, 11, 11] has median 11; upper half [17, 17, 23] has median 17. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

12. Mean 13/2 min; MAD 5/2 min

Mean = 39/6 = 13/2. Absolute deviations: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2. Their sum divided by 6 is 5/2. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

13. 6 observations; minutes; median 8, IQR 2; mean 8, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 8 and half medians give IQR 2. Mean is 48/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

14. 6 observations; minutes; median 16, IQR 6; mean 16, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 16 and half medians give IQR 6. Mean is 96/6; averaging distances gives MAD 5. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

15. 6 observations; minutes; median 14, IQR 6; mean 14, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 14 and half medians give IQR 6. Mean is 84/6; averaging distances gives MAD 5. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

16. 6 observations; minutes; median 27/2, IQR 5; mean 27/2, MAD 25/6. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 27/2 and half medians give IQR 5. Mean is 81/6; averaging distances gives MAD 25/6. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

17. First median 9; second 13 min; both IQR 2 min.

Adding 4 shifts every position and both quartiles by 4. The median increases by 4; Q3 - Q1 remains 2.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

18. 13 students; stacks at 0, 1, 2, 3 have 1, 3, 4, 5 dots.

Place one dot per student, not one per distinct value. Total dots = 1 + 3 + 4 + 5 = 13. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

19. B, D - What is the reading time in minutes of each of the students?; How does reading time vary among the students?

Different students can have different reading time. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

20. A, D - The median is the average of the 3rd and 4th ordered values.; The median is 9 minutes.

Six ordered observations have middle positions 3 and 4, so median = (8 + 10)/2 = 9.

6.SP.B.5c | Multiple select